Métodos Numéricos ICCD412

U4 Interpolación y Ajuste de Curvas

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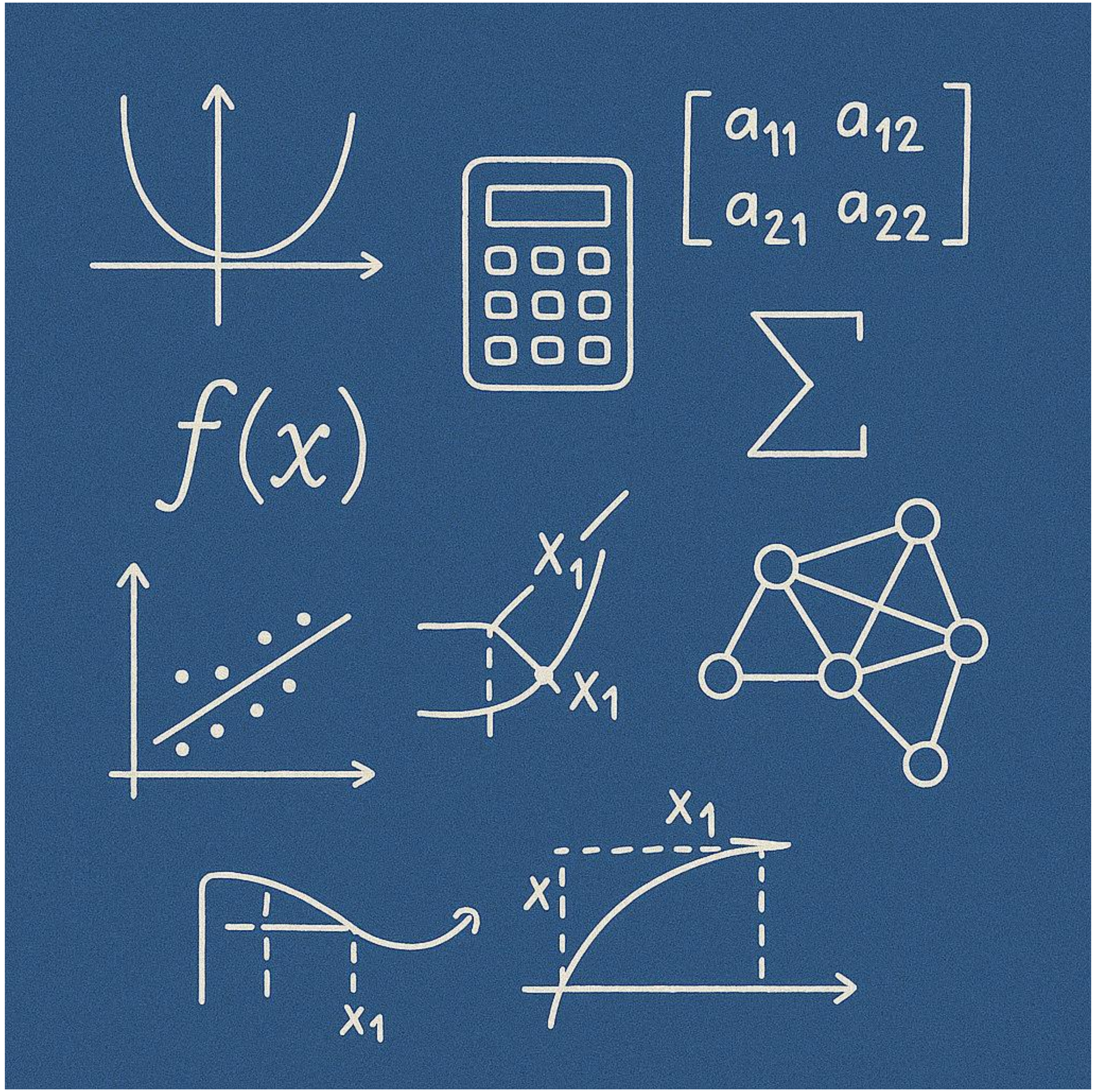

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- Derivada por regla de la cadena
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Derivada por regla del producto

Si tienes $f(x) = u(x) \times v(x)$, entonces:

$$f'(x) = u'(x) \times v(x) + u(x) \times v'(x)$$

$$f(x) = x^2 \sin(x)$$
$$f'(x) = u'(x) \times v(x) + u(x) \times v'(x)$$
$$f'(x) = 2x \sin(x) + x^2 \cos(x)$$

Derivada por regla de la cadena

Si tienes $f(x) = g\big(h(x)\big)$, entonces:

$$f'(x) = g'\big(h(x)\big) \times h'(x)$$

$$f(x) = \underbrace{\cos(\overbrace{x^2}^{\text{interior}})}_{\text{exterior}}$$

$$f'(x) = g'\big(h(x)\big) \times h'(x)$$
$$f'(x) = -\sin(x^2) \times 2x$$

Derivada parcial

$$f(x,y) = x^2y$$

$$\frac{\partial f}{\boldsymbol{\partial x}} = \underbrace{\frac{\partial}{\boldsymbol{\partial x}}\boldsymbol{x^2}y}_{\text{Trata a } y \text{ como una constante}} = \boldsymbol{2x}y$$

$$\frac{\partial f}{\boldsymbol{\partial y}} = \underbrace{\frac{\partial}{\boldsymbol{\partial y}}x^2\boldsymbol{y}}_{\text{Trata a } x \text{ como una constante}} = x^2 * \boldsymbol{1}$$

Derivada parcial

$$\frac{\overbrace{\partial f}^{\substack{\text{un pequeño cambio en} \\ \text{el resultado de la función}}}}{\underbrace{\partial x}_{\text{un pequeño cambio en } x}}$$

Se usa en lugar de "d" en la notación usual de $\frac{df}{dx}$ para enfatizar que esta es una derivada parcial.

Función multivariable

Indica cuál variable de entrada se cambia ligeramente.

Derivada parcial

$$f(x,y) = \frac{1}{5}(x^2 - 2xy) + 3$$

$$\frac{\partial f}{\boldsymbol{\partial x}} = \frac{1}{5}\frac{\partial}{\boldsymbol{\partial x}}\left(\boldsymbol{x^2} - 2\boldsymbol{x}y\right) + 3 = \frac{1}{5}(\boldsymbol{2x} - 2 * \boldsymbol{1} * y)$$

$$\frac{\partial f}{\boldsymbol{\partial y}} = \frac{1}{5}\frac{\partial}{\boldsymbol{\partial y}}(x^2 - 2x\boldsymbol{y}) + 3 = \frac{1}{5}(-2x * \boldsymbol{1})$$

Derivada parcial

Cálculo por regla de la potencia: $f(x, y) = x^n y^m$

$$\frac{\partial f}{\boldsymbol{\partial x}} = \frac{\partial}{\boldsymbol{\partial x}} \boldsymbol{x^n} y^m = \boldsymbol{n x^{n-1}} y^m$$

$$\frac{\partial f}{\boldsymbol{\partial y}} = \frac{\partial}{\boldsymbol{\partial y}} x^n \boldsymbol{y^m} = \boldsymbol{m} x^n \boldsymbol{y^{m-1}}$$

Derivada parcial

Cálculo por regla del producto: $f(x,y) = g(x)h(y)$

$$\frac{\partial f}{\boldsymbol{\partial x}} = \frac{\partial}{\boldsymbol{\partial x}}\boldsymbol{g(x)}h(y) = \boldsymbol{g'(x)}h(y)$$

$$\frac{\partial f}{\boldsymbol{\partial y}} = \frac{\partial}{\boldsymbol{\partial y}}g(x)\boldsymbol{h(y)} = g(x)\boldsymbol{h'(y)}$$

Derivada parcial

Cálculo por regla de la cadena: $f(x, y) = g(u, v)$ y $u = u(x, y), v = v(x, y)$

$$\frac{\partial f}{\partial x} = \frac{\partial g}{\partial u}\frac{\partial u}{\partial x} + \frac{\partial g}{\partial v}\frac{\partial v}{\partial x}$$

$$\frac{\partial f}{\partial y} = \frac{\partial g}{\partial u}\frac{\partial u}{\partial y} + \frac{\partial g}{\partial v}\frac{\partial v}{\partial y}$$

Derivada parcial

$$E(a_0, a_1) = \sum_{i=1}^{n} [y_i - (a_1 x_i + a_0)]^2$$

$$\frac{\delta E}{\delta a_0} = ?$$

$$\frac{\delta E}{\delta a_1} = ?$$

Derivada parcial

$$E(a_0, a_1) = \sum_{i=1}^{n} [y_i - (a_1 x_i + a_0)]^2$$

$$\frac{\partial E}{\partial a_0} = \sum_{i=1}^{n} \frac{\partial g}{\partial e_i} \frac{\partial e_i}{\partial a_0}$$

$$\frac{\partial E}{\partial a_1} = \sum_{i=1}^{n} \frac{\partial g}{\partial e_i} \frac{\partial e_i}{\partial a_1}$$

Splines Cúbicos

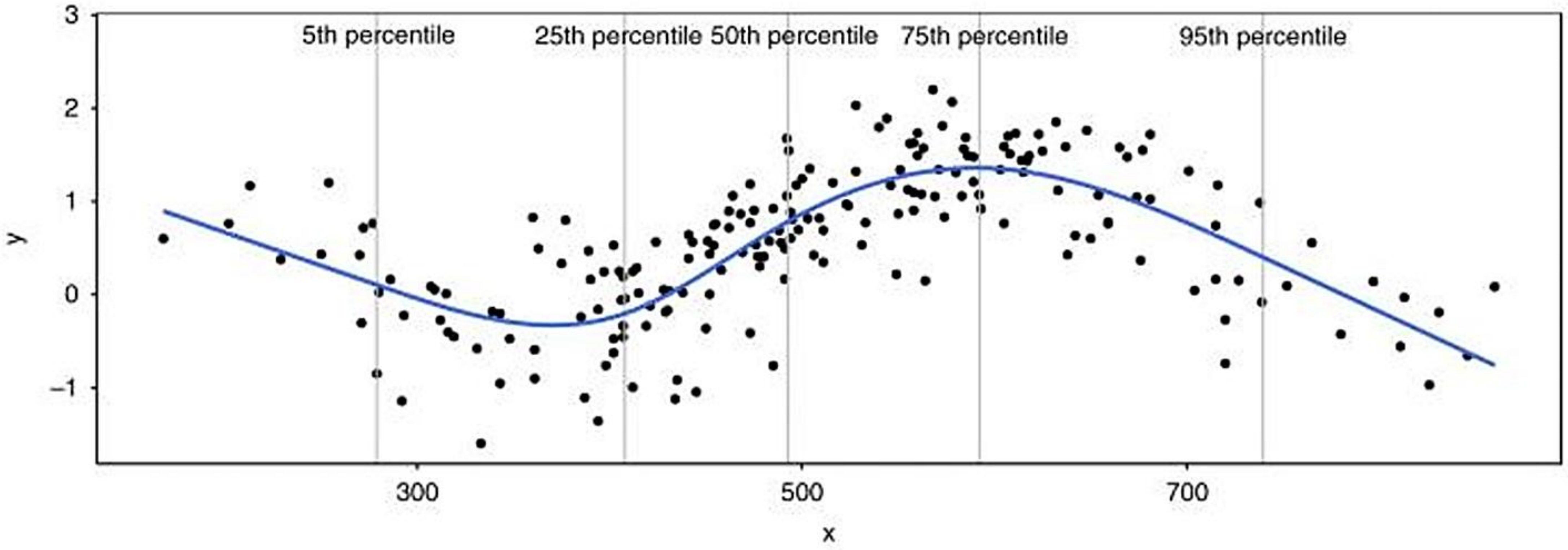
Si tienes $f(x) = u(x) \times v(x)$, entonces:

$$f'(x) = u'(x) \times v(x) + u(x) \times v'(x)$$

$$f(x) = x^2 \sin(x)$$
$$f'(x) = u'(x) \times v(x) + u(x) \times v'(x)$$
$$f'(x) = 2x \sin(x) + x^2 \cos(x)$$

Splines Cúbicos

Función cúbica por partes que interpola un conjunto de puntos de datos y garantiza suavidad en los puntos de datos



Splines Cúbicos

Dados los puntos $\{x_0, y_0, x_1, y_1, x_2, y_2, \dots x_n, y_n\}$ de la función $\boldsymbol{f(x)}$ entre $[\boldsymbol{a}, \boldsymbol{b}]$

- $a = x_0 < x_1 < \cdots < x_n = b$
  - (ordenados)
- $\boldsymbol{S(x)}$ es un spline cúbico
  - $S_j(x) = a_j + b_j(x - x_j) + c_j(x - x_j)^2 + d_j(x - x_j)^3$
- $S_j(x)$ polinomio de 3er grado en $[x_j, x_{j+1}]$, y $j = 0,1, \dots n - 1$

Splines Cúbicos

- El polinomio j coincide en los puntos dados:
  - $S_j(x_j) = y_j$
  - $S_j(x_{j+1}) = y_{j+1}$
- Los polinomios se intersecan en (x_j, y_j):
  - $S_j(x_{j+1}) = S_{j+1}(x_{j+1})$
- La derivada y la 2da derivada coinciden entre polinomios contiguos:
  - $S_j'(x_{j+1}) = S_{j+1}'(x_{j+1})\ S_j''(x_{j+1}) = S_{j+1}''(x_{j+1})$

Splines Cúbicos

Condiciones de borde o frontera

- *Frontera natural*
  - $S_0''(x_0) = S_{n-1}''(x_n) = 0$

- *Frontera condicionada*
  - **2 entradas adicionales* $\{B_0, B_n\}$
  - $S_0'(x_0) = f'(x_0),\ S_{n-1}'(x_n) = f'(x_n)$

Spline cúbico natural

Para construir el spline cúbico interpolante S para la función f, definido en los números $x_0 < x_1 < \cdots < x_n$, que satisfacen $S''(x_0) = S''(x_n) = 0$:

ENTRADA $n; x_0, x_1, \ldots, x_n; a_0 = f(x_0), a_1 = f(x_1), \ldots, a_n = f(x_n)$.

SALIDA a_j, b_j, c_j, d_j para $j = 0, 1, \ldots, n-1$.
(*Nota:* $S(x) = S_j(x) = a_j + b_j(x - x_j) + c_j(x - x_j)^2 + d_j(x - x_j)^3$ *para* $x_j \leq x \leq x_{j+1}$.)

Paso 1 Para $i = 0, 1, \ldots, n-1$ haga $h_i = x_{i+1} - x_i$.

Paso 2 Para $i = 1, 2, \ldots, n-1$ haga

$$\alpha_i = \frac{3}{h_i}(a_{i+1} - a_i) - \frac{3}{h_{i-1}}(a_i - a_{i-1}).$$

Paso 3 Determine $l_0 = 1$; (*Los pasos 3, 4 y 5 y parte del paso 6 resuelven un sistema lineal tridiagonal con un método descrito en el algoritmo 6.7.*)

$\mu_0 = 0$;
$z_0 = 0$.

Paso 4 Para $i = 1, 2, \ldots, n-1$
haga $l_i = 2(x_{i+1} - x_{i-1}) - h_{i-1}\mu_{i-1}$;
$\mu_i = h_i / l_i$;
$z_i = (\alpha_i - h_{i-1}z_{i-1})/l_i$.

Paso 5 Haga $l_n = 1$;
$z_n = 0$;
$c_n = 0$.

Paso 6 Para $j = n-1, n-2, \ldots, 0$
haga $c_j = z_j - \mu_j c_{j+1}$;
$b_j = (a_{j+1} - a_j)/h_j - h_j(c_{j+1} + 2c_j)/3$;
$d_j = (c_{j+1} - c_j)/(3h_j)$.

Paso 7 SALIDA $(a_j, b_j, c_j, d_j$ para $j = 0, 1, \ldots, n-1)$;
PARE.

Spline cúbico condicionado

Para construir el spline cúbico interpolante S para la función f definida en los números $x_0 < x_1 < \cdots < x_n$, que satisfacen $S'(x_0) = f'(x_0)$ y $S'(x_n) = f'(x_n)$:

ENTRADA n; $x_0, x_1, \ldots, x_n$; $a_0 = f(x_0), a_1 = f(x_1), \ldots, a_n = f(x_n)$; $FPO = f'(x_0)$; $FPN = f'(x_n)$.

SALIDA a_j, b_j, c_j, d_j para $j = 0, 1, \ldots, n-1$.
(*Nota:* $S(x) = S_j(x) = a_j + b_j(x-x_j) + c_j(x-x_j)^2 + d_j(x-x_j)^3$ *para* $x_j \le x \le x_{j+1}$.)

Paso 1 Para $i = 0, 1, \ldots, n-1$ haga $h_i = x_{i+1} - x_i$.

Paso 2 Haga $\alpha_0 = 3(a_1 - a_0)/h_0 - 3FPO$;
$\alpha_n = 3FPN - 3(a_n - a_{n-1})/h_{n-1}$.

Paso 3 Para $i = 1, 2, \ldots, n-1$

haga $\alpha_i = \dfrac{3}{h_i}(a_{i+1} - a_i) - \dfrac{3}{h_{i-1}}(a_i - a_{i-1})$.

Paso 4 Haga $l_0 = 2h_0$; (*Los pasos 4, 5 y 6 y parte del paso 7 resuelven un sistema lineal tridiagonal con un método descrito en el algoritmo 6.7.*)
$\mu_0 = 0.5$;
$z_0 = \alpha_0/l_0$.

Paso 5 Para $i = 1, 2, \ldots, n-1$
haga $l_i = 2(x_{i+1} - x_{i-1}) - h_{i-1}\mu_{i-1}$;
$\mu_i = h_i/l_i$;
$z_i = (\alpha_i - h_{i-1}z_{i-1})/l_i$.

Paso 6 Haga $l_n = h_{n-1}(2 - \mu_{n-1})$;
$z_n = (\alpha_n - h_{n-1}z_{n-1})/l_n$;
$c_n = z_n$.

Paso 7 Para $j = n-1, n-2, \ldots, 0$
haga $c_j = z_j - \mu_j c_{j+1}$;
$b_j = (a_{j+1} - a_j)/h_j - h_j(c_{j+1} + 2c_j)/3$;
$d_j = (c_{j+1} - c_j)/(3h_j)$.

Paso 8 SALIDA (a_j, b_j, c_j, d_j para $j = 0, 1, \ldots, n-1$);
PARE. ■

Dados los puntos $x = [0, 1, 2],\ y = [-5, -4, 3]$

a) Determine el spline cúbico con frontera natural

b) Determine el spline cúbico con frontera condicionada

2 splines cúbicos

- $S_0(x) = a_0 + b_0(x - x_0) + c_0(x - x_0)^2 + d_0(x - x_0)^3$
- $S_1(x) = a_1 + b_1(x - x_1) + c_1(x - x_1)^2 + d_1(x - x_1)^3$

[0,1] y [1,2]

- 8 incógnitas → 8 ecuaciones

(# puntos -1)* 4

Coincidencia con los puntos de datos.

- $S_0(x) = a_0 + b_0(x-0) + c_0(x-0)^2 + d_0(x-0)^3$
- $S_1(x) = a_1 + b_1(x-1) + c_1(x-1)^2 + d_1(x-1)^3$

- $S_0(x_0) = y_0$

$$S_0(x_0) = a_0 + b_0(x_0-0) + c_0(x_0-0)^2 + d_0(x_0-0)^3 = y_0$$

$$S_0(x_0) = a_0 + b_0(0-0) + c_0(0-0)^2 + d_0(0-0)^3 = y_0$$

$$S_0(x_0) = a_0 = y_0 = -5 \qquad (1)$$

- $S_0(x_1) = y_1$

$$S_0(x_1) = a_0 + b_0(x_1-0) + c_0(x_1-0)^2 + d_0(x_1-0)^3 = y_1$$

$$S_0(x_1) = a_0 + b_0(1-0) + c_0(1-0)^2 + d_0(1-0)^3 = y_1$$

$$S_0(x_1) = -5 + b_0 + c_0 + d_0 = -4$$

$$S_0(x_1) = b_0 + c_0 + d_0 = 1 \qquad (2)$$

Coincidencia con los puntos de datos.

- $S_0(x) = a_0 + b_0(x-0) + c_0(x-0)^2 + d_0(x-0)^3$
- $S_1(x) = a_1 + b_1(x-1) + c_1(x-1)^2 + d_1(x-1)^3$

- $S_1(x_1) = y_1$

$$S_1(x_1) = a_1 + b_1(x_1-1) + c_1(x_1-1)^2 + d_1(x_1-1)^3 = y_1$$

$$S_1(x_1) = a_1 + b_1(1-1) + c_1(1-1)^2 + d_1(1-1)^3 = y_1$$

$$S_1(x_1) = a_1 = y_1 = -4 \qquad (3)$$

- $S_1(x_2) = y_2$

$$S_1(x_2) = a_1 + b_1(x_2-1) + c_1(x_2-1)^2 + d_1(x_2-1)^3 = y_2$$

$$S_1(x_2) = a_1 + b_1(2-1) + c_1(2-1)^2 + d_1(2-1)^3 = y_2$$

$$S_1(x_2) = -4 + b_1 + c_1 + d_1 = 3$$

$$S_1(x_2) = b_1 + c_1 + d_1 = 7 \qquad (4)$$

Continuidad de la primera derivada.

$$S_0'(x_1) = S_1'(x_1) \qquad\qquad S_j'(x) = b_j + 2c_j(x - x_j) + 3d_j(x - x_j)^2$$

$$b_0 + 2c_0(x_1 - x_0) + 3d_0(x_1 - x_0)^2 = b_1 + 2c_1(x_1 - x_1) + 3d_1(x_1 - x_1)^2$$

$$b_0 + 2c_0(1 - 0) + 3d_0(1 - 0)^2 = b_1 + 2c_1(1 - 1) + 3d_1(1 - 1)^2$$

$$b_0 + 2c_0 + 3d_0 = b_1 \qquad (5)$$

Continuidad de la segunda derivada.

$$S_0''(x_1) = S_1''(x_1) \qquad\qquad S_j''(x) = 2c_j + 6d_j(x - x_j)$$

$$2c_0 + 6d_0(x_1 - x_0) = 2c_1 + 6d_1(x_1 - x_1)$$

$$2c_0 + 6d_0(1 - 0) = 2c_1 + 6d_1(1 - 1)$$

$$2c_0 + 6d_0 = 2c_1 \qquad (6)$$

Condiciones de frontera

Frontera Natural

$$S_0''(x_0) = S_{n-1}''(x_n) = 0 \qquad S_j''(x) = 2c_j + 6d_j(x - x_j)$$

$$S_0''(x_0) = S_{2-1}''(x_2) = 0$$

$$2c_0 + 6d_0(x_0 - x_0) = 2c_1 + 6d_1(x_2 - x_1) = 0$$

$$2c_0 + 6d_0(0 - 0) = 2c_1 + 6d_1(2 - 1) = 0$$

$$2c_0 = 0 \qquad (7a)$$

$$2c_1 + 6d_1 = 0 \qquad (8a)$$

Condiciones de frontera

Frontera Condicionada

2 entradas adicionales $\{B_0, B_n\}$

$S_0'(x_0) = f'(x_0) = B_0,\ S_{n-1}'(x_n) = f'(x_n) = B_n$

$S_j'(x) = b_j + 2c_j\left(x - x_j\right) + 3d_j\left(x - x_j\right)^2$

$$S_0'(x_0) = b_0 + 2c_0(x_0 - x_0) + 3d_0(x_0 - x_0)^2 = B_0$$
$$S_0'(x_0) = b_0 + 2c_0(0 - 0) + 3d_0(0 - 0)^2 = B_0$$

$b_0 = B_0$ (7b)

$$S_{2-1}'(x_2) = b_1 + 2c_1(x_2 - x_1) + 3d_1(x_2 - x_1)^2 = B_n$$
$$S_{1-1}'(x_2) = b_1 + 2c_1(2 - 1) + 3d_1(2 - 1)^2 = B_n$$

$b_1 + 2c_1 + 3d_1 = B_n$ (8b)

$$a_0 = -5 \quad (1)$$

$$b_0 + c_0 + d_0 = 1 \quad (2)$$

$$a_1 = -4 \quad (3)$$

$$b_1 + c_1 + d_1 = 7 \quad (4)$$

$$b_0 + 2c_0 + 3d_0 = b_1 \quad (5)$$

$$2c_0 + 6d_0 = 2c_1 \quad (6)$$

$$2c_0 = 0 \quad (7a)$$

$$2c_1 + 6d_1 = 0 \quad (8a)$$

$$a_0 = -5 \quad (1)$$

$$b_0 + c_0 + d_0 = 1 \quad (2)$$

$$a_1 = -4 \quad (3)$$

$$b_1 + c_1 + d_1 = 7 \quad (4)$$

$$b_0 + 2c_0 + 3d_0 = b_1 \quad (5)$$

$$2c_0 + 6d_0 = 2c_1 \quad (6)$$

$$b_0 = B_0 \quad (7b)$$

$$b_1 + 2c_1 + 3d_1 = B_n \quad (8b)$$

$$B_0 = 1$$

$$B_n = -1$$

Frontera Natural

$S_0(x) = -5 - 0.5x + 1.5x^3$

$S_1(x) = -4 + 4(x-1) + 4.5(x-1)^2 - 1.5(x-1)^3$

$$\boldsymbol{S}(x) = \begin{cases} -5 - 0.5x + 1.5x^3 & , 0 \leq x \leq 1 \\ -4 + 4(x-1) + 4.5(x-1)^2 - 1.5(x-1)^3 & , 1 < x \leq 2 \end{cases}$$

Frontera Condicionada

$S_0(x) = -5 + x - 5x^2 + 5x^3$

$S_1(x) = -4 + 6(x-1) + 10(x-1)^2 - 9(x-1)^3$

$$\boldsymbol{S}(x) = \begin{cases} -5 + x - 5x^2 + 5x^3 & ,0 \leq x \leq 1 \\ -4 + 6(x-1) + 10(x-1)^2 - 9(x-1)^3 & ,1 < x \leq 2 \end{cases}$$

Dados los puntos $x = [-2, -1, 1, 3]$, $y = [3, 1, 2, -1]$

a) Determine el spline cúbico con frontera natural

b) Determine el spline cúbico con frontera condicionada

$$B_0 = 1$$

$$B_n = -1$$

Temas a codificar

Spline cúbico natural

Spline cúbico condicionado

OBRIGADO
gracias
どうも
ARIGATO
hvala
DANK U
takk
MERSI
merci
謝謝
obrigado
danke schön
köszi
PALDIES
muchas gracias
ありがとう
TEŞEKKÜR EDERIM
MOLTE GRAZIE
GO RAIBH MAITH AGAT
danke
grazas
GRAZZI
THANKS
qujan
THANK YOU
благодаря
TAK
どうも
TAK
asante
muchas gracias
vielen dank
PALDIES
DANK U
OBRIGADO
mesi
danke
감사합니다
köszi
благодаря
grazie
DZIEKI
TACK
Gràcies
спасибо
MULTUMESC
多謝
TEŞEKKÜR EDERIM
muchas gracias
obrigado
NA GODE